

Same answers as lista7_resolucao.pdf, in the format of the instructor's keys: arrow chains instead of paragraphs, results highlighted, traps as # notes. Every step is shown. Lower-case letters are log-deviations from an efficient steady state. Calibration (Benigno p. 515, fn. 24): $\alpha = 0,66$, $\sigma = 0,5$, $\eta = 0,2$, $\theta = 8$, $\rho = 2\%$. SR = short run, LR = long run.

Here σ is the EIS (scaled by $s_c = C/Y$: $\sigma = s_c \tilde{\sigma}$); in Kurlat σ is curvature and $1/\sigma$ the EIS. Opposite conventions.

(1) NK vs New Classical (NC) vs original Keynesian (K)

Summary: NK = NC method (optimisation, rational expectations) + K friction (prices don't clear in the SR), plus monopolistic competition and an interest-rate instrument.

(i) Prices, expectations, monetary policy

- K: rigidity assumed, adaptive/no expectations $\rightarrow$ stable, exploitable Phillips curve $\rightarrow$ money has lasting real effects.
- NC: flexible prices, RE $\rightarrow$ only surprises move y (Lucas) $\rightarrow$ systematic policy ineffective (Sargent–Wallace).
- NK: RE + share α of firms keeps a price p^e set *before* the shock. The AS curve follows in four steps:

1. Adjusting firm: $\tilde{P} = (1 + \mu) \frac{W}{A} \rightarrow \frac{\tilde{P}}{P} = (1 + \mu) \frac{W/P}{A}$; household: $\frac{W}{P} = \frac{v_l(L)}{u_c(C)} = L^\eta C^{\tilde{\sigma}^{-1}}$.
2. Flexible prices ($\tilde{P} = P$) define Y_n : $(1 + \mu) \frac{L_n^\eta C_n^{\tilde{\sigma}^{-1}}}{A} = 1$. Divide step 1 by this and take logs: $\tilde{p} - p = \eta(y - y_n) + \tilde{\sigma}^{-1}(c - c_n)$.
3. $y = s_c c + g \rightarrow c - c_n = \frac{y - y_n}{s_c}$, and $\tilde{\sigma}^{-1} = s_c \sigma^{-1} \rightarrow \tilde{p} - p = (\sigma^{-1} + \eta)(y - y_n)$ (real marginal cost).
4. Index: $p = \alpha p^e + (1 - \alpha) \tilde{p} \rightarrow \tilde{p} - p = \frac{p - \alpha p^e}{1 - \alpha} - p = \frac{\alpha}{1 - \alpha}(p - p^e)$. Equate with step 3:

$$p - p^e = \kappa(y - y_n), \quad \kappa = \frac{(1 - \alpha)(\sigma^{-1} + \eta)}{\alpha} \quad \kappa = \frac{0,34 \cdot 2,2}{0,66} = \frac{0,748}{0,66} = 1,133$$

- Same equation as Lucas, but p^e is a *posted price*, not a forecast $\rightarrow$ **known policy reacting after p^e is set still moves y** .
LR: all firms reset $\rightarrow y = \bar{y}_n$ (vertical LR AS) $\rightarrow$ money neutral.
- Evidence: 1970s stagflation (keep RE); announced disinflations still cost output (need a friction that fools no one); micro prices last ≈ 3 quarters $\rightarrow \alpha \approx 0,66$.
- Implication: $\partial \kappa / \partial \alpha < 0$ (α in the denominator) $\rightarrow$ more rigidity $\rightarrow$ flatter AS; $\alpha \rightarrow 0 \rightarrow \kappa \rightarrow \infty \rightarrow$ classical.

Same curve as Lucas (Benigno p. 508), not the forward-looking NKPC.

(ii) Market structure

- K: unmodelled / rigid nominal wage with competitive firms $\rightarrow$ firms on falling $MPL \rightarrow$ countercyclical real wage (not in data).

- NC: perfect competition $\rightarrow$ nobody sets a price $\rightarrow$ nobody to be sticky.
- NK: Dixit–Stiglitz demand $Y(j) = \left(\frac{P(j)}{P}\right)^{-\theta} (C+G)$, cost $\frac{W}{A}$ per unit. Profit $\Pi = P(j)^{1-\theta} P^\theta (C+G) - \frac{W}{A} P(j)^{-\theta} P^\theta (C+G)$.
FOC: $(1-\theta)P(j)^{-\theta} + \theta \frac{W}{A} P(j)^{-\theta-1} = 0 \rightarrow$ multiply by $P(j)^{\theta+1}$: $(\theta-1)P(j) = \theta \frac{W}{A} \rightarrow$

$$P(j) = \underbrace{\frac{\theta}{\theta-1}}_{\text{mark-up}} \frac{W}{A} \quad (\theta > 1).$$
- Consequences: (1) firms *set* prices; (2) $P > MC \rightarrow$ a firm stuck at p^e gains on each extra unit $\rightarrow$ serves demand: y demand-determined in the SR; (3) mark-up = wedge $\rightarrow y_n < y_e$ (Q3).
- Menu cost: firm's loss from not adjusting is 2nd order (envelope: $\partial \Pi / \partial P(j) = 0$ at the optimum); aggregate output effect is 1st order (Mankiw 1985).

(iii) Aggregate demand and the instrument

- K: IS (consumption function) + LM (money demand); CB sets M ; slope via real balances ($P \uparrow \rightarrow M/P \downarrow \rightarrow i \uparrow \rightarrow I \downarrow$).
- NC: $MV = PY$; M is the instrument and the nominal shock.
- NK: AD from the Euler equation, CB sets i , no LM. Derivation:

1. Euler: $u_c(C) = \beta(1+i) \frac{P}{\bar{P}} u_c(\bar{C})$, with $u_c = C^{-\tilde{\sigma}-1}$.
2. Logs ($\ln \beta = -\rho$, $\ln(1+i) \approx i$): $-\tilde{\sigma}^{-1}c = -\rho + i + p - \bar{p} - \tilde{\sigma}^{-1}\bar{c} \rightarrow c = \bar{c} - \tilde{\sigma} [i - (\bar{p} - p) - \rho]$.
3. Multiply by s_c and use $y - g = s_c c$, $\bar{y} - \bar{g} = s_c \bar{c}$, $\sigma = s_c \tilde{\sigma}$, $\bar{y} = \bar{y}_n$:

$$y = \bar{y}_n + (g - \bar{g}) - \sigma [i - (\bar{p} - p) - \rho] \quad \left. \frac{dp}{dy} \right|_{AD} = -\frac{1}{\sigma} < 0$$

- Slope: $p \uparrow \rightarrow (\bar{p} - p) \downarrow$ (expected inflation) $\rightarrow r = i - (\bar{p} - p) \uparrow \rightarrow C \downarrow \rightarrow y \downarrow$.
- Evidence: CBs target short rates; money demand unstable. Implication: transmission via r ; expectations $(\bar{y}_n, \bar{p})$ shift demand today; ZLB possible.

No LM curve: never explain AD's slope with money demand.

(iv) Microfoundations and welfare

- K: postulated equations, ad hoc loss. NC: micro-founded but competitive $\rightarrow$ efficient (1st welfare theorem) $\rightarrow$ nothing to stabilise.
- NK: micro-founded *with* distortions (μ , sticky p) $\rightarrow$ loss derived from utility (2nd-order approximation around the efficient SS, Benigno eq. 33, after Woodford 2003):

$$L = \frac{1}{2}(y - y_e)^2 + \frac{\theta}{2\kappa}(p - p^e)^2$$
- Reading: target y_e (not y_n); only surprises $p - p^e$ cost (dispersion between adjusters and frozen firms); weight $\frac{\theta}{\kappa} = \frac{\theta\alpha}{(1-\alpha)(\sigma^{-1}+\eta)} \rightarrow \uparrow$ in α and in θ .
- Evidence: Lucas critique (1970s model failures) $\rightarrow$ deep parameters $(\beta, \tilde{\sigma}, \eta, \theta, \alpha)$.

The approximation itself is not derived in the article; the course takes eq. 33 as given.

(2) AS-AD. Start E : $i = \rho$, $y = y_n = y_e = 0$, $p = p^e = \bar{p}$, $g = \bar{g}$.

Closed form (used in both items). Put AS, $p = p^e + \kappa(y - y_n)$, into AD with $\bar{p} = p^e$:

$$\begin{aligned} y &= \bar{y}_n - \sigma [i - p^e + p^e + \kappa(y - y_n) - \rho] = \bar{y}_n - \sigma(i - \rho) - \sigma\kappa(y - y_n) \\ y - y_n &= (\bar{y}_n - y_n) - \sigma(i - \rho) - \sigma\kappa(y - y_n) \\ (1 + \sigma\kappa)(y - y_n) &= \sigma \left[\underbrace{\rho + \sigma^{-1}(\bar{y}_n - y_n)}_{\equiv r_n} - i \right] \end{aligned}$$

$$y - y_n = \frac{\sigma}{1 + \sigma\kappa}(r_n - i), \quad p - p^e = \kappa(y - y_n) \quad \sigma\kappa = 0,5 \cdot 1,133 = 0,567, \quad 1 + \sigma\kappa = 1,567$$

At E : $r_n = \rho = i \rightarrow y - y_n = 0$. ✓

(a) **Temporary** $\mu \uparrow \Rightarrow$ AS shifts up/left; AD does not move

Which curve. From (15), $y_n = \frac{(1 + \eta)a + \sigma^{-1}g - \mu}{\sigma^{-1} + \eta} \rightarrow dy_n = -\frac{d\mu}{\sigma^{-1} + \eta} < 0$; from (19), $\mu \notin y_e \rightarrow dy_e = 0$.

AS anchor $(p^e, y_n) \rightarrow (p^e, y'_n) \rightarrow$ AS shifts left; at fixed y it shifts up by

$$dp|_y = -\kappa dy_n = \frac{(1 - \alpha)(\sigma^{-1} + \eta)}{\alpha} \cdot \frac{d\mu}{(\sigma^{-1} + \eta)} = \frac{1 - \alpha}{\alpha} d\mu$$

AD: $\mu \notin$ AD; temporary $\rightarrow \bar{\mu}, \bar{y}_n$ unchanged $\rightarrow$ AD fixed.

Effects. $di = 0$, $d\bar{y}_n = 0 \rightarrow dr_n = -\sigma^{-1}dy_n > 0$. Closed form:

$$\begin{aligned} d(y - y_n) &= \frac{\sigma}{1 + \sigma\kappa} dr_n = -\frac{dy_n}{1 + \sigma\kappa} > 0 \\ dy &= dy_n + d(y - y_n) = dy_n \left(1 - \frac{1}{1 + \sigma\kappa} \right) \end{aligned}$$

$$dy = \frac{\sigma\kappa}{1 + \sigma\kappa} dy_n < 0, \quad dp = \kappa d(y - y_n) = -\frac{\kappa}{1 + \sigma\kappa} dy_n > 0 \quad \text{stagflation}$$

Mechanism: adjusters want a bigger margin $\rightarrow p \uparrow \rightarrow (\bar{p} - p) \downarrow \rightarrow r \uparrow \rightarrow C \downarrow \rightarrow y \downarrow$ (along AD); frozen firms keep serving at $p^e \rightarrow y$ falls less than y_n .

Numbers, $d\mu = 5\%$:

- $dy_n = -5/2,2 = -2,273\%$; $dr_n = 2,273/0,5 = 4,545$ pp $\rightarrow r_n = 6,55\%$
- $dy = \frac{0,567}{1,567}(-2,273) = 0,362 \cdot (-2,273) = -0,82\%$
- $dp = \frac{1,133 \cdot 2,273}{1,567} = \frac{2,576}{1,567} = +1,64\%$; AS shift $= \frac{0,34}{0,66} \cdot 5 = 2,58\%$
- Gaps: $y - y_n = \frac{2,273}{1,567} = +1,45\% > 0$ (why $p \uparrow$); $y - y_e = dy - dy_e = -0,82\% < 0$ (why welfare $\downarrow$)

LR: every firm resets $\rightarrow y = \bar{y}_n = 0$, $p = \bar{p} = p^e$.

Two gaps, opposite signs. Demand falls along AD. A permanent $\mu \uparrow$ would also lower $\bar{y}_n \rightarrow$ AD down ($dr_n = 0$).

(b) $i \downarrow \Rightarrow$ AD shifts up by $|di|$ (right by $\sigma|di|$); AS does not move

Which curve. Solve AD for p : $\sigma[i - \bar{p} + p - \rho] = \bar{y}_n - y \rightarrow p = \bar{p} + \rho - i + \frac{\bar{y}_n - y}{\sigma}$
 $\rightarrow$ fixed y : $dp = -di > 0$ (up); fixed p : $dy = -\sigma di > 0$ (right). $i \notin$ AS; y_n depends on a, g, μ ; p^e predetermined $\rightarrow$ AS fixed.

Effects. $dy_n = 0$, $dr_n = 0$. Closed form:

$$dy = d(y - y_n) = -\frac{\sigma}{1 + \sigma\kappa} di > 0, \quad dp = \kappa dy > 0$$

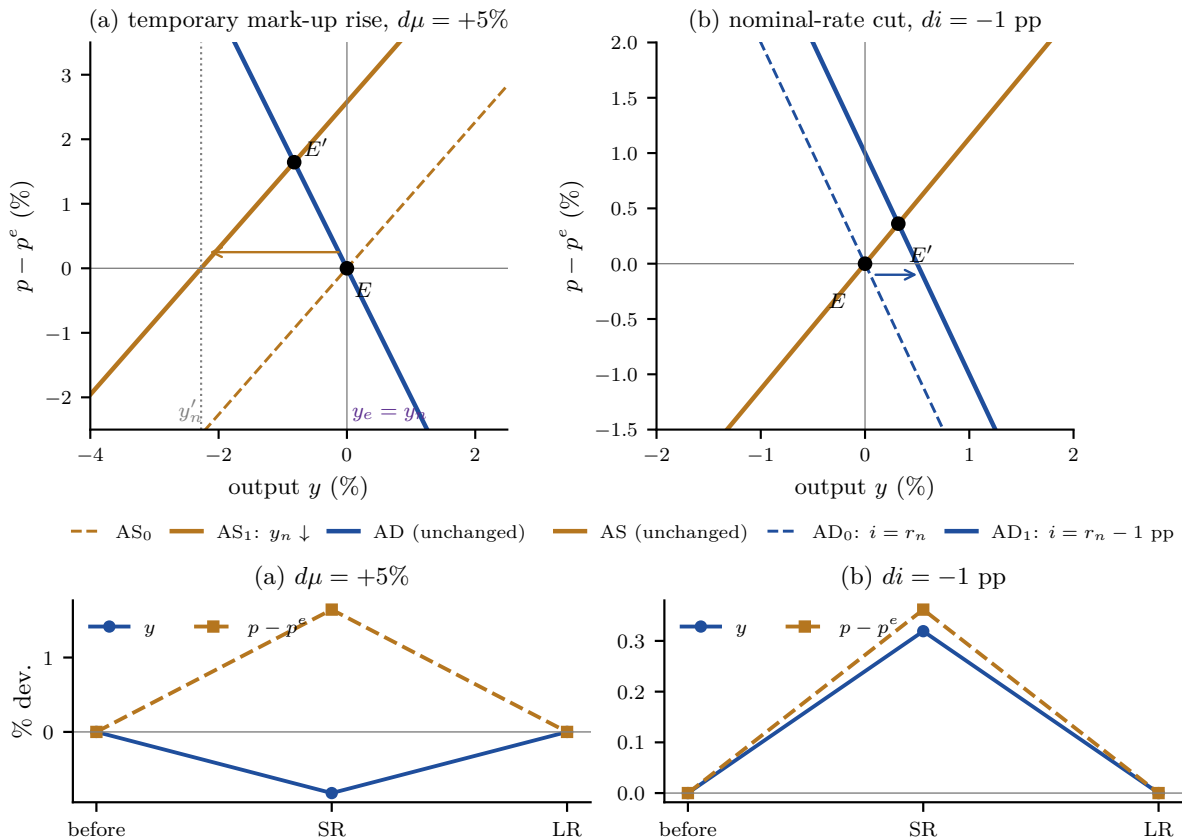
Mechanism: $i \downarrow \rightarrow r \downarrow$ ($\bar{p}$ anchored) $\rightarrow C \uparrow \rightarrow y \uparrow$ (sticky firms serve) $\rightarrow$ hours $\uparrow \rightarrow W/P$ required $\uparrow \rightarrow MC \uparrow \rightarrow$ adjusters raise $p \uparrow \rightarrow r$ partly back up.

Numbers, $di = -1$ pp:

- AD: up 1 pp, right $0,5 \cdot 1 = 0,5$ pp
- $dy = \frac{0,5 \cdot 1}{1,567} = +0,32\%$; $dp = 1,133 \cdot 0,319 = +0,36\%$
- Share of AD shift reaching y : $\frac{dy}{\sigma|di|} = \frac{1}{1 + \sigma\kappa} = \frac{1}{1,567} = 64\%$
- $dy_n = dy_e = 0 \rightarrow y - y_n = y - y_e = +0,32\%$ (overheating)

LR: money neutral, $y \rightarrow \bar{y}_n$; only $\bar{p}$ could move.

No LM / liquidity effect: CB sets i directly. $\alpha \uparrow \rightarrow \kappa \downarrow \rightarrow$ more y , less p .



(3) y_n vs y_e

- y_n : flexible prices, *decentralised* economy (market power and taxes kept) $\rightarrow$ AS anchor; $y - y_n$ drives **prices**.
- y_e : *planner*, $\max u(C) - v(L)$ s.t. $Y = C + G$, $Y = AL$; $y - y_e$ drives **welfare**. Neither depends on α .

Planner. Substitute both constraints: $\max_Y u(Y - G) - v(Y/A) \rightarrow u_c(Y - G) - \frac{1}{A}v_l(Y/A) = 0 \rightarrow \frac{v_l}{u_c} = A$ (MRS = MRT).

Market. Every firm resets: $P = (1 + \mu)\frac{W}{A} \rightarrow \frac{W}{P} = \frac{A}{1 + \mu}$; household $\frac{v_l}{u_c} = \frac{W}{P} \rightarrow \frac{v_l}{u_c} = \underbrace{\frac{A}{1 + \mu}}_{\text{wedge}}$.

Logs. $v_l = L^\eta$, $L = Y/A \rightarrow \ln v_l = \eta(y - a)$; $u_c = C^{-\tilde{\sigma}^{-1}}$, $c = \frac{y-g}{s_c}$, $\tilde{\sigma}^{-1} = s_c\sigma^{-1} \rightarrow \ln u_c = -\sigma^{-1}(y - g) \rightarrow \ln \frac{v_l}{u_c} = \eta(y - a) + \sigma^{-1}(y - g)$.

$$\text{planner: } \eta(y_e - a) + \sigma^{-1}(y_e - g) = a \rightarrow (\sigma^{-1} + \eta)y_e = (1 + \eta)a + \sigma^{-1}g$$

$$\text{market: } \eta(y_n - a) + \sigma^{-1}(y_n - g) = a - \mu \rightarrow (\sigma^{-1} + \eta)y_n = (1 + \eta)a + \sigma^{-1}g - \mu$$

Subtract: $(\sigma^{-1} + \eta)(y_n - y_e) = -\mu \rightarrow$

$$y_n - y_e = -\frac{\mu}{\sigma^{-1} + \eta} \quad \mu = 5\% : -\frac{5}{2,2} = -2,27\%$$

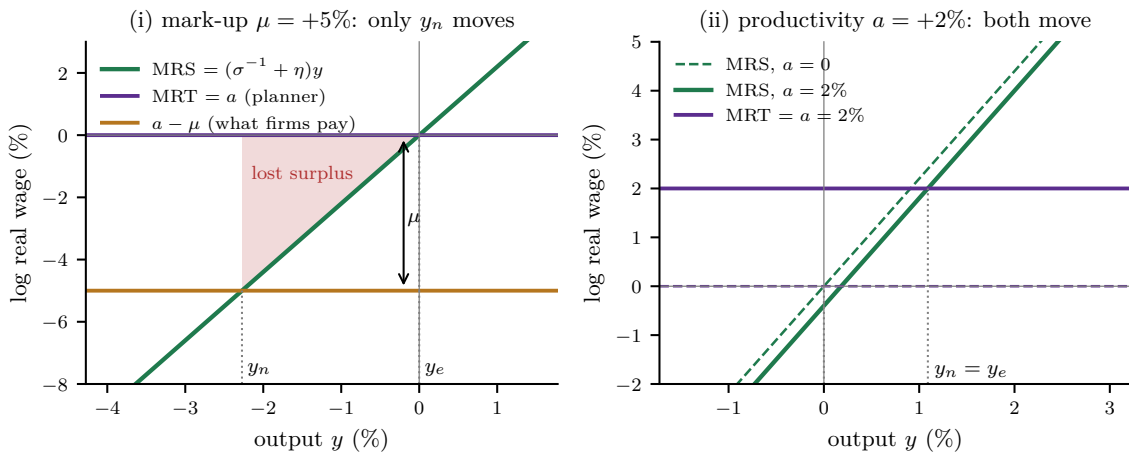
Why they differ. $P > MC \rightarrow W/P = A/(1 + \mu) < A = MPL \rightarrow$ household stops at $MRS = A/(1 + \mu) \rightarrow L, Y$ too low. Lost surplus = triangle between $MRT = a$ and $\ln MRS$ from y_n to y_e : height μ (gap at y_n), base $y_e - y_n = \frac{\mu}{\sigma^{-1} + \eta} \rightarrow \frac{1}{2}\mu \cdot \frac{\mu}{\sigma^{-1} + \eta} = \frac{1}{2}\frac{\mu^2}{\sigma^{-1} + \eta} = \frac{1}{2} \cdot \frac{0,05^2}{2,2} = 0,057\%$ of output. Quadratic $\rightarrow$ loss in $(y - y_e)^2$.

Why a temporary $\mu \uparrow$ moves only y_n . $y_n = y_n(a, g, \mu)$, $y_e = y_e(a, g)$: the planner's problem has no prices, market power or taxes

$$dy_n = -\frac{d\mu}{\sigma^{-1} + \eta} < 0, \quad dy_e = 0 \rightarrow \text{inefficient shock} \rightarrow \text{trade-off}$$

Contrast $a \uparrow$: enters both $\rightarrow dy_n = dy_e = \frac{1+\eta}{\sigma^{-1}+\eta}da = \frac{1,2}{2,2} \cdot 2 = 1,09\%$ for $da = 2\%$ (no trade-off).

$\# \alpha$ appears in neither: stickiness explains $y \neq y_n$, not $y_n \neq y_e$.



$$(4) \quad \min \mathcal{L} = \phi_y(y - y^*)^2 + \phi_p(p - p^e)^2 \text{ s.t. } p - p^e = \kappa(y - y_n)$$

Shape. $\mathcal{L} \geq 0$, $= 0$ only at the bliss point (y^*, p^e) . Quadratic $\rightarrow$ symmetric (booms as bad as recessions: the CB *stabilises*) and convex ($\partial \mathcal{L} / \partial y = 2\phi_y(y - y^*)$ is 0 at the target and grows with the miss)

$\rightarrow$ spread the miss across both targets. Iso-loss $\phi_y(y - y^*)^2 + \phi_p(p - p^e)^2 = \bar{\mathcal{L}}$: ellipses, axes $\propto (1/\sqrt{\phi_y}, 1/\sqrt{\phi_p})$. ϕ_p/ϕ_y = conservatism (hawk vs dove). Only surprises cost (dispersion). Welfare version: $\phi_y = \frac{1}{2}$, $\phi_p = \frac{\theta}{2\kappa}$, $y^* = y_e$ (eq. 33).

Why subject to AS. One instrument i , and i appears only in AD $\rightarrow$ moving i slides AD *along* AS $\rightarrow$ every point of AS is reachable, no other point is $\rightarrow$ AS = feasible set (“budget line”), slope κ = price of output in surprises. AD is not a constraint: for any (y, p) on AS there is an i putting AD through it.

Solve (substitute the constraint, no Lagrangian):

$$\begin{aligned} \mathcal{L}(y) &= \phi_y(y - y^*)^2 + \phi_p \kappa^2 (y - y_n)^2 \\ \mathcal{L}'(y) &= 2\phi_y(y - y^*) + 2\phi_p \kappa^2 (y - y_n) = 0 \quad \mathcal{L}''(y) = 2(\phi_y + \phi_p \kappa^2) > 0 \text{ (minimum)} \end{aligned}$$

Divide by 2, write $\kappa(y - y_n) = p - p^e$:

$$\phi_y(y - y^*) + \kappa \phi_p(p - p^e) = 0 \quad (\text{targeting rule})$$

Solve $\mathcal{L}' = 0$ for y : $(\phi_y + \phi_p \kappa^2)y = \phi_y y^* + \phi_p \kappa^2 y_n$. Let $D \equiv \phi_y + \phi_p \kappa^2$, $\Delta \equiv y^* - y_n$:

$$\begin{aligned} y^o &= \frac{\phi_y y^* + \phi_p \kappa^2 y_n}{D}, \quad y^o - y_n = \frac{\phi_y}{D} \Delta, \quad y^o - y^* = -\frac{\phi_p \kappa^2}{D} \Delta \\ p^o - p^e &= \kappa(y^o - y_n) = \frac{\kappa \phi_y}{D} \Delta \\ \mathcal{L}^o &= \phi_y \frac{\phi_p^2 \kappa^4}{D^2} \Delta^2 + \phi_p \kappa^2 \frac{\phi_y^2}{D^2} \Delta^2 = \frac{\phi_y \phi_p \kappa^2 (\phi_p \kappa^2 + \phi_y)}{D^2} \Delta^2 = \frac{\phi_y \phi_p \kappa^2}{\phi_y + \phi_p \kappa^2} (y^* - y_n)^2 \end{aligned}$$

Share into prices, relative to strict output ($p - p^e = \kappa \Delta$): $\frac{p^o - p^e}{\kappa \Delta} = \frac{\phi_y}{D}$. Welfare weights: $\phi_p \kappa^2 = \frac{\theta}{2\kappa} \kappa^2 = \frac{\theta \kappa}{2} \rightarrow \frac{\phi_y}{D} = \frac{1/2}{1/2 + \theta \kappa / 2} = \frac{1}{1 + \theta \kappa}$; rule $\frac{1}{2}(y - y_e) + \kappa \frac{\theta}{2\kappa}(p - p^e) = 0 \rightarrow (y - y_e) + \theta(p - p^e) = 0$ (eq. 34).

Implementing rate. From the closed form of Q2: $y^o - y_n = \frac{\sigma}{1 + \sigma \kappa}(r_n - i^o) \rightarrow i^o = r_n - \frac{1 + \sigma \kappa}{\sigma}(y^o - y_n)$.

Numbers, $d\mu = 5\%$, welfare weights, $y^* = y_e = 0$:

- $\Delta = 0 - (-2,273) = 2,273$; $\theta \kappa = 8 \cdot 1,133 = 9,07$; share $= 1/10,07 = 0,099$
- $y^o - y_n = 0,099 \cdot 2,273 = 0,226 \rightarrow y^o = -2,273 + 0,226 = -2,05\%$; $p^o - p^e = 1,133 \cdot 0,226 = +0,26\%$
- $\phi_p = \frac{8}{2 \cdot 1,133} = 3,53$, $\phi_p \kappa^2 = 4,53$, $D = 5,03 \rightarrow \mathcal{L}^o = \frac{0,5 \cdot 4,53}{5,03} \cdot 2,273^2 = 0,450 \cdot 5,17 = 2,33$
- $i^o = 6,55 - \frac{1,567}{0,5} \cdot 0,226 = 6,55 - 0,71 = 5,84\%$ (vs 2% doing nothing; vs $r_n = 6,55\%$ for strict price stability)
- Strict output ($y = 0$): $p - p^e = 1,133 \cdot 2,273 = 2,58\%$, $i = 6,55 - 3,13 \cdot 2,273 = -0,58\%$ (a cut; zero bound ignored)

Sign of the response (vs doing nothing, $i = \rho$; with $y^* = y_e = 0$, so $r_n - \rho = -y_n/\sigma = \Delta/\sigma$).
 $i^o > \rho \iff \frac{1+\sigma\kappa}{\sigma} \frac{\phi_y}{D} \Delta < r_n - \rho = \frac{\Delta}{\sigma} \iff \phi_y(1 + \sigma\kappa) < \phi_y + \phi_p\kappa^2 \iff \underbrace{\frac{\phi_y}{\phi_p\kappa}}_{|\text{IT slope}|} < \underbrace{\frac{1}{\sigma}}_{|\text{AD slope}|}$.

Welfare: $\frac{\phi_y}{\phi_p\kappa} = \frac{1}{\theta} = 0,125 < 2 \rightarrow \text{raise}$. IT flatter than AD $\rightarrow$ raise i ; steeper $\rightarrow$ cut: the sign is a property of the mandate.

Policy meaning.

- $\Delta = 0$ (a, g shocks move $y_n = y_e$ together) $\rightarrow y^o = y^*, p^o = p^e, \mathcal{L}^o = 0, i = r_n$: divine coincidence.
- $\Delta \neq 0$ (μ shocks) $\rightarrow \mathcal{L}^o > 0$: trade-off, split $\frac{\phi_y}{D}$ into prices.
- CB picks a point *on* AS, cannot move AS $\rightarrow$ cannot change $y_n - y_e \rightarrow$ taxes / competition policy.
- Trade-off only SR and only with sticky prices: $\alpha \rightarrow 0 \rightarrow \kappa \rightarrow \infty \rightarrow \frac{\phi_y}{D} \rightarrow 0 \rightarrow y^o \rightarrow y_n, p^o \rightarrow p^e$. Can't hold $y > y_n$ every period: with rational p^e , $E[p - p^e] = 0 \rightarrow E[y - y_n] = 0$ (Kydland–Prescott).

AS is the constraint, not AD. κ is private pricing technology, not a policy choice.

